

100 Days of Linux - Day 033

Title: Counting Lines, Words, and Characters with `wc`

Today's Goal

Learn how to count lines, words, characters, and bytes in files using the `wc` command.

Why This Matters

Developers and data engineers use `wc` to quickly inspect log files, datasets, documentation, and source code.

Command of the Day: `wc`

Syntax: `wc filename`

Common Options:

`wc -l file` (count lines)

`wc -w file` (count words)

`wc -m file` (count characters)

`wc -c file` (count bytes)

Examples:

`wc employees.txt`

`wc -l sales.csv`

Beginner Lesson

The `wc` command summarizes text files. It helps you quickly understand the size of a file without opening it. This is especially useful for checking CSV files, reports, and logs.

Practice Questions

1. Create a file named `notes.txt` with at least 10 lines. Count the number of lines using `wc`.
2. Count the number of words in `notes.txt`.
3. Count the number of characters in `notes.txt`.
4. Create `employees.csv` with at least 5 records and count its lines using `wc -l`.
5. Display your current working directory after completing today's tasks.

Hints

Use Nano to create files. Try `wc` first, then experiment with `-l`, `-w`, `-m`, and `-c`.

Mini Challenge

Create `report.txt` with 20 lines. Display its line, word, character, and byte counts. Compare the outputs of `wc`, `wc -l`, `wc -w`, `wc -m`, and `wc -c`.

Common Mistakes

Confusing characters with bytes. For plain English text they are

often the same, but they can differ for some Unicode characters.

Did You Know?

The `wc` command can also read input from other commands through a pipe, such as `grep` or `cat`.

Pro Tip

Use `wc -l` to quickly estimate the number of records in a CSV file (remember that one line may be the header).

Answer Key (Instructor Only)

Q1: `wc -l notes.txt`

Q2: `wc -w notes.txt`

Q3: `wc -m notes.txt`

Q4: `wc -l employees.csv`

Q5: `pwd`

Homework

Create three different text files and practice counting lines, words, characters, and bytes using every `wc` option.

Next Day Preview

Tomorrow you'll learn the `cut` command to extract specific columns and fields from text and CSV files.